Business_only3 Project Handbook & Architecture Guide

This PDF is a complete hybrid handbook and interview-ready guide for the Business_only3 platform. It explains the system end-to-end, including architecture, security decisions, and production readiness.

1. Project Purpose and Vision

Business_only3 was built to simulate a real-world enterprise system. The goal was to create a secure, role-based platform with internal communication, admin governance, and production-style architecture.

2. Roles and Access Control

The platform supports Admin, Manager, Gospel, and Staff roles. Access is strictly enforced using session isolation and identity helpers.

3. Folder Architecture

/admin – internal administration area
/includes – shared session and identity logic
/ajax – asynchronous operations
/attachment – secure file uploads
/config.php – centralized database configuration

4. Authentication Flow

Authentication uses PHP sessions with role-bound logic. Admin and user sessions are completely isolated.

5. Identity Layer

Identity helpers abstract role checks and session validation, preventing duplication and improving security.

6. Database Architecture

Core tables include users, admin, roles, feedback, notification, and contacts. Each table enforces ownership and role-based constraints.

7. Messaging Architecture

All chat messages use a unified feedback table with channel separation (user_user, user_admin, admin_admin).

8. Notification System

Notifications are filtered by role and receiver. Admins can see all, while users see only relevant alerts.

9. Security Decisions

Passwords use password_hash(). Legacy hashes are upgraded automatically. All database access uses prepared statements.

10. Common Issues Solved

Invalid parameter errors were fixed by aligning placeholders. Session collisions were solved with isolated namespaces. Avatar caching issues were fixed using timestamped URLs.

11. Admin Profile & Avatar System

Admin avatars are stored as BLOBs and served securely via avatar_admin.php, avoiding public file exposure.

12. Chat UI Enhancements

Chat headers dynamically display peer name and role, e.g., Chat with John K (Staff).

13. Interview Explanation

This project demonstrates real-world patterns: role separation, secure authentication, modular architecture, and scalability.

14. Production Hardening Checklist

HTTPS enforcement
Rate limiting
Audit logging
Database backups
Monitoring and alerting

15. Scaling Strategy

Sessions can move to Redis, chat can be separated into a service, and queues can be added for notifications.

16. Final Summary

Business_only3 is a production-style system built with enterprise principles. It is suitable for demonstrations, interviews, and real-world expansion.

End of Handbook